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1 1) Core choice: what is an “area quantum”?

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1.1 You

Please search for knowledge related “Principal Differential Geometric Algebra” and “compute using Grassmann.jl, Cartan.jl” as explained in a book of Michael Reed and in the github project under <https://github.com/chakravala>.

Question 1: Are you aware in the sense of “ready to apply” of all math related to “Differential Geoemtri Algebra”?

Question 2: Can you apply this the foundations of Burkhard Heims theory?

1.2 ChatGPT

I'll look up the relevant book/project context (Michael Reed + chakravala / Grassmann.jl / Cartan.jl), then answer both questions directly in terms of what I can apply now and what would still need formalization.

I found the key references and package claims. Next I'll verify a concise description of Burkhard Heim's theory context so I can answer Q2 precisely.

Thought for 17s

Yes — I'm now oriented to the exact ecosystem you mean:

Grassmann.jl is presented as a Grassmann-Clifford-Hodge multilinear differential geometric algebra framework with exterior/regressive/inner/geometric products, Hodge star, boundary operators, and compile-time tensor-algebra typing via DirectSum.jl / AbstractTensors.jl. [GitHub](#)

Cartan.jl is presented as a Julia framework for tensor fields / frame bundles / manifolds, aiming at executable differential geometric algebra and geometric PDEs on grids/meshes/simplicial decompositions, with TensorField constructions and references to Reed's DGA texts/books. [GitHub](#)

The GitHub/Makie issue confirms the 2025/2026 “Principal Differential Geometric Algebra: compute using Grassmann.jl, Cartan.jl” book context and its relation to Grassmann/Cartan/-Makie examples. [GitHub](#)

Q1

No, not in the literal sense of “all math, ready-to-apply” for every statement in that body of work.

What I am ready to apply (at a useful level):

exterior algebra / Clifford (geometric) algebra identities

differential forms, Hodge duality, wedge/interior products
connections, curvature forms, Lie derivatives/brackets (standard formulations)
manifold/tangent-bundle coordinate constructions
translation of many of these objects into Grassmann.jl / Cartan.jl-style computational workflows

What is not safe to claim without your concrete target chapter/formulation:

exact coverage of all Reed-specific definitions, conventions, and notation choices

every package-specific API nuance/version detail

every derivation in a 545-page book (especially if it introduces nonstandard terminology or custom identities)

So: ready to work with it seriously, yes; omniscient over all details, no.

Q2

Yes, I can help apply differential geometric algebra machinery to the foundational mathematics of Burkhard Heim-related formulations — but with an important condition:

I need a precise formalization of which “Heim theory” layer you mean (original Heim, later extensions, a specific author’s reconstruction, mass formula sector, geometric axioms, etc.).

Heim theory is not mainstream / not broadly accepted in contemporary physics, and available descriptions are often mixed with speculative or secondary interpretations. Wikipedia+1

What I can do well:

recast claims into explicit math objects (manifolds, bundles, forms, metrics, connections, operators)

check internal consistency (dimensions, index structure, covariance, algebra closure)

derive a computable prototype in Julia using Grassmann.jl / Cartan.jl

separate:

mathematical formalism

computational implementation

empirical/physical claims (which require independent evidence)

Practical way to proceed

If you want, send one small target (even rough notes), e.g.:

“Heim’s space dimensional decomposition”

“mass operator / metric ansatz”

“field equations in notation X”

“one chapter theorem / derivation from Reed”

and I can convert it into a DG-algebra / Cartan-form formulation plus a Julia skeleton (types, basis, operators, test identities).

Sources

1.3 You

You are right Heim Theory has several components and extensions. I would like to focus on the smallest possible scale, where Heim defines some area quanta, these elements are clearly defined by Heim and most extension do not cover their details. But to develop Heim's theory we should start with such area quanta and their detailed description. Are these rectangular, triangle shape or hexagonal. Are they bi-vectors and if yes in what space. I would like to assume rigid elements whose area/size might change over time. Heim had discussed some spheres as a starting point in time and then these spheres expand. Some sort energy function related to the curvature of these spheres is a global energy balance. We can assume that these elements are very small (Planck length). We can assume that there is an energy related to angles between such elements (only the hinges can freely move). Such a geometric structure has already a big number of constraints (3-diagonal like sparse matrix form). At this stage we have no fields, no particle and no possibility to generate some ad hoc curvature. To have some bending at one place we have to move a infinite amount of elements or we have to assume a curved boundary condition. Please try to generate a theory according to Heim ideas build from the area quanta. Assume that movement of curved ensemble take no energy, but their is causality speed limit. To move such an ensemble, you have to bend some elements in front of the structure and relax some elements at the end of the structure. Please keep in mind that quantum theory's wave function and the square of its value might be discovered as simple functions of curvature, where the square values are related to energy and mass.

1.4 ChatGPT

I'll build a minimal geometric model from area quanta only (kinematics + constraints + energy + causal update rule), then show how a wavefunction-like quantity can emerge from curvature amplitudes.

Thought for a few seconds

Excellent direction. Starting from area quanta (not particles/fields) is the right "smallest-scale" move if the goal is to make Heim-like ideas mathematically testable.

Below is a minimal constructive theory you can actually develop. It is not a claim about Heim's exact original equations; it is a Heim-inspired foundation built from area elements, hinges, curvature, and causal propagation.

1) Core choice: what is an "area quantum"? Recommendation

Treat each area quantum as a bivector-valued oriented plaquette:

geometrically: a small rigid 2-cell (triangle is best)

algebraically: a bivector $B \in \Lambda^2(T_x M)$

metric size: $|B| \sim \ell_P^2$ (Planck-area scale)

shape: fixed combinatorial type, variable metric embedding

Why triangles (best minimal choice)

Triangles are topologically stable building blocks for arbitrary surfaces/manifolds (simplicial complex).

Local curvature at hinges is standard (Regge-like).

No ambiguity in planarity of a 2-cell.

Hexagons/rectangles are convenient for lattices, but triangles are universal.

Alternative choices

Rectangles: easier indexing, but less natural under arbitrary curvature (shear ambiguity).

Hexagons: good for near-isotropic tilings in 2D, but not universal simplicial primitives.

Conclusion: use triangular plaquettes as primitive, allow coarse-grained hexagonal patterns to emerge.

2) What space are the bivectors in?

This is crucial.

Minimal answer

Each area quantum is a bivector in the local tangent space:

$$B_f \in \Lambda^2(T_x M)$$

where f labels a face (triangle) and x is its location (or cell center).

If you want a stricter algebraic implementation:

choose a local frame e_a ,

represent

$$B_f = \frac{1}{2} B_f^{ab} e_a \wedge e_b. \text{ Which dimension?}$$

You can keep this general at first:

Kinematic model: M is a 3D spatial manifold and time is an update parameter.

Covariant model: M is 4D spacetime and plaquettes may be spacelike/timelike.

Heim-inspired extension: if you later need extra dimensions, you can enlarge $T_x M$ to n -D without changing the plaquette machinery.

Best starting point: Use 3D space + discrete time. Then each area quantum is a spatial bivector in $\Lambda^2(\mathbb{R}^3)$, which is dual to a normal vector. This makes intuition easy.

3) Minimal kinematics: rigid faces, flexible hinges

Your idea is strong: rigid elements + hinge motion.

Define a simplicial complex K with:

vertices V

edges E

triangular faces F

Each face $f \in F$:

has fixed rest area $A_{0,f} \sim \ell_P^2$ (or nearly fixed)

carries orientation and bivector B_f

may allow small area dilation $A_f(t)$

Each pair of adjacent faces sharing edge e :

has a dihedral angle θ_e

This gives a discrete geometry with no continuous metric field assumed.

4) Curvature from hinges (discrete and local)

This is the natural place to define curvature without “ad hoc curvature.”

Hinge curvature variable

For each edge e , define a curvature/bending variable

$$\kappa_e \equiv \theta_e - \theta_{0,e}$$

where $\theta_{0,e}$ is the reference angle (flat or vacuum value).

$\kappa_e = 0$: local vacuum shape

$\kappa_e \neq 0$: stored bending / curvature excitation

This is exactly the right object if the only free motions are hinge motions.

5) Energy functional (local, geometric, no fields yet)

You asked for:

area/size may change over time

energy from curvature of expanding spheres / global balance

energy from hinge angles

curved ensemble motion can be “free” except causal limit

A good minimal energy is:

$$E = E_{\text{area}} + E_{\text{bend}} + E_{\text{compat}} + E_{\text{boundary}} \quad \text{(a) Area term (small dilation allowed) } E_{\text{area}} = \sum_{f \in F} \frac{K_A}{2} \left(\frac{A_f - A_{0,f}}{A_{0,f}} \right)^2 A_{0,f}$$

K_A : area stiffness

if you want rigid area quanta, take $K_A \rightarrow \infty$

$$\text{(b) Hinge bending term (main curvature energy) } E_{\text{bend}} = \sum_{e \in E} \frac{K_\theta}{2} \kappa_e^2 L_e$$

L_e edge length weight

this is the discrete analog of curvature-squared elasticity / geometric energy

(c) Compatibility/closure constraints

To prevent impossible local shapes, include penalties (or exact constraints):

face closure

edge length consistency

orientation compatibility

discrete Bianchi-like closure (at a vertex/cell)

Symbolically:

$$E_{\text{compat}} = \lambda_C \sum (\text{constraint violations})^2 \quad (\text{d}) \text{ Boundary / global curvature condition}$$

You mentioned curved boundary conditions. Yes: curvature can be injected by boundary data.

$$E_{\text{boundary}} = \text{penalty enforcing prescribed boundary curvature/shape}$$

This avoids needing “infinite elements moved” in a finite simulation; the boundary carries the global geometry.

6) Vacuum and “expanding spheres” (Heim-inspired seed state)

Your idea of spheres expanding from an initial state can be formalized as a reference family of embeddings.

Define a vacuum background as a discrete spherical ensemble

At time t , let the reference configuration be approximately a sphere of radius $R(t)$:

faces tiled on a triangulated sphere

reference angles $\theta_{0,e}(t)$ chosen to match that sphere

reference areas $A_{0,f}(t)$ may scale with expansion

Then energy is measured relative to this moving vacuum:

$$\kappa_e(t) = \theta_e(t) - \theta_{0,e}(t)$$

This is powerful because:

“expansion of spheres” is built into vacuum kinematics

excitations are deviations from expansion, not the expansion itself

You also proposed:

movement of curved ensemble takes no energy, but there is a causality speed limit

That suggests a gauge-like degeneracy:

certain collective deformations are zero modes of E

they carry information/configuration change but no energy cost (or negligible cost)

dynamics constrained by local update speed c

7) Dynamics: causal hinge updates (no continuum fields required)

Define discrete time steps $t_n \rightarrow t_{n+1}$.

State variables

At each step:

face bivectors B_f^n

edge curvatures κ_e^n

possibly conjugate momenta p_e^n

Local update rule (causal)

An edge/face can update only using neighbors within one graph step:

$\text{update}(e, n + 1) = F(\text{neighborhood of } e \text{ at } n)$

with a speed bound:

no influence beyond distance $c \Delta t$ (converted to graph distance)

This gives your causality rule without assuming standard fields.

Transport of a curved ensemble

Your picture is exactly a soliton-like curvature packet on the hinge network:

front: increase curvature (bend elements)

rear: decrease curvature (relax elements)

packet translates with low net energy change if profile is near a zero mode

This is analogous to defect transport in lattice systems.

8) Why “infinite elements must move” appears — and how to avoid it

You correctly noticed a deep issue:

In a globally constrained rigid network, local bending can force long-range rearrangement.

This is not a bug; it is a compatibility consequence.

Three ways to handle it:

A. Curved boundary condition (your proposal)

Prescribe boundary curvature/shape, solve interior compatibility.

physically: universe/background geometry is not flat infinity

mathematically: well-posed finite-domain problem

B. Expanding vacuum reference

Measure only deviations from a curved/expanding reference $\theta_{0,e}(t)$.

then many “global moves” become vacuum-preserving gauge motions

C. Topological defects

Allow sparse defects/disclinations/dislocations.

localized curvature can then exist without moving everything

these defects may later become particle-like excitations

This is likely the bridge from pure geometry to matter-like structures.

9) Wavefunction as curvature amplitude (your key insight)

This is a very interesting and plausible route.

You suggested:

wavefunction amplitude from curvature

square relates to energy/mass

A clean formulation is:

Curvature complex amplitude

Define a local complex scalar from geometric invariants:

$$\psi(x, t) = \alpha \kappa(x, t) + i \beta \pi(x, t)$$

where:

κ : coarse-grained curvature amplitude (from hinge angles)

π : conjugate “curvature flux” / rate / phase-like variable

α, β : scaling constants

Then define density:

$$\rho(x, t) = |\psi(x, t)|^2$$

Interpretations:

$\rho_E \propto |\psi|^2$: local geometric excitation energy density

$\rho_M \propto |\psi|^2 / c^2$: effective mass density after coarse-graining

This is not yet quantum mechanics, but it gives the square-law density naturally.

How phase can emerge geometrically

Phase may come from:

orientation/rotation of bivectors

holonomy around loops

discrete connection angle (parallel transport defect)

Then interference-like behavior can emerge when multiple curvature transport paths recombine and phases add before squaring.

That gives a route toward a wavefunction interpretation from geometric holonomy + curvature amplitudes.

10) Minimal formal theory (compact version)

Here is a concise axiomatic version you can iterate on.

Axiom A1 — Discrete geometric substrate

Space at the fundamental scale is a simplicial complex K of oriented triangular area quanta f , each represented by a bivector $B_f \in \Lambda^2(T_x M)$.

Axiom A2 — Rigid plaquettes, flexible hinges

Plaquettes are metrically rigid to first order ($A_f \approx A_{0,f}$), while relative orientation changes across shared edges are dynamical through dihedral angles θ_e .

Axiom A3 — Curvature as hinge deviation

Local curvature excitation is encoded by hinge variables $\kappa_e = \theta_e - \theta_{0,e}$, relative to a reference vacuum geometry (possibly expanding and curved).

Axiom A4 — Geometric energy

Energy is a functional of area strain, hinge curvature, and compatibility constraints:

$E[B, \theta] = \sum_f U_A(A_f) + \sum_e U_\theta(\kappa_e) + \sum_c U_C(\text{closure/compatibility})$ Axiom A5 — Causal local dynamics

The substrate evolves by local update rules with finite propagation speed c , preserving orientation and compatibility constraints up to allowed defect creation rules.

Axiom A6 — Curvature packets and transport

Localized curvature patterns propagate by front-bending/rear-relaxation; translation may occur along near-zero-energy modes while remaining causally bounded.

Axiom A7 — Emergent amplitude

A coarse-grained complex amplitude ψ is defined from curvature and its conjugate geometric flux; $|\psi|^2$ is proportional to local excitation energy density and effective inertial density.

11) What this buys you (important)

This framework gives you, before particles/fields:

a kinematic substrate

a curvature-based energy

a causal update law

a natural square-law density

a path to defects = matter-like excitations

a path to holonomy phase = wave behavior

That matches your goal very well.

12) Immediate next step (concrete and testable)

Build the smallest nontrivial model:

Model M0: 2D triangulated shell in 3D (discrete time)

Triangular mesh approximating a sphere

Fixed edge lengths or stiff edges

Hinge angle energy

Expanding reference radius $R(t)$

Inject one localized curvature packet

Observe transport under local causal rule

Measure:

total energy

packet velocity (bounded)

dispersion

interference of two packets

$|\psi|^2$ -like density from curvature and rate

This is the right scale to test your hypothesis before adding any “field” language.

13) Julia / Grassmann.jl / Cartan.jl angle (how to encode)

Even before full simulation, you can define the algebra cleanly.

Grassmann.jl use

represent oriented local basis and bivectors $e_i \wedge e_j$

compute wedge/inner/geometric products

encode orientation and local frame transforms

Cartan.jl use

discrete/coarse differential objects, frame fields, connections (as you scale up)

later: curvature/holonomy in a bundle-like formulation

For the first prototype, you may only need:

a mesh structure

hinge angles

local frames

bivector bookkeeping

Then gradually add Cartan-style connection/holonomy objects.

14) One key caution

To stay disciplined (and make this scientific), keep three layers separate:

Layer 1: exact geometry (definitions, constraints, energy)

Layer 2: emergent phenomenology (packets, wave amplitude, mass density)

Layer 3: Heim correspondence (which parts map to Heim’s original concepts)

This prevents accidental overfitting to interpretations.

If you want, next I can give you a formal M0 specification in math + a Julia prototype skeleton (mesh, hinge curvature, energy, causal update, curvature-to- ψ mapping) in a style compatible with Grassmann.jl.

1.5 You

Great starting point. I would like you to focus only “Layer 1” the geometric constraints and postpone discussion related “Layer 2” and “Layer 3” to quantum area change over time and connections to quantum theory. I think these can be derived if the basic theory is settled. You can also remove everything related to bending of area quantum elements, these can be considered to be so elementary that no other properties exist. Related to the coordinate systems and dimensions I have a different view. The time axis should be consider the same as the space dimension at the beginning, while later the movement of curvature in the time direction might be identified with electromagnetic effects. We should assume that for the simple existence of an curvature in space it must also exist in the time dimension and must move in the time cone to have a prolonged existence. Please note that photons in this

theory might no “move” at all, because their proper time stands still. Please re-check details of Heim’s discussion how particle mass is created by combination of several dimensions. Because we are using “differential geometric algebra” it might be that some terms from quantum mechanics like “phase space, configuration space, DoF” and our highly constraint movement and Heim’s higher dimension are related to some “dual space” construct where solutions would be quite easy to derive using GA concepts of Hodge operator.

1.6 ChatGPT

I’ll tighten this to a pure Layer-1 geometric substrate with your revised assumptions (time initially symmetric with space, no plaquette bending, only existence/compatibility constraints), and I’ll keep Heim-specific claims conservative.

Thought for 13s

Agreed. This is the right refinement.

We should freeze the theory at Layer 1 and make it purely geometric:

no quantum interpretation yet

no fields/particles yet

no bending/deformation of the area quanta themselves

only existence, adjacency, orientation, compatibility, and causal persistence

And your revised symmetry assumption is strong:

time starts as a dimension on equal footing with space

persistent localized curvature must have a component that “threads” the time direction (a world-tube / world-sheet condition)

later asymmetry (e.g. electromagnetic interpretation) can emerge, but not be assumed

That is consistent with secondary summaries of Heim theory emphasizing a quantized smallest area (“metron”), a 6D quantized lattice-like geometry, and the idea that different physical structures correspond to different dimensional deformation classes, including ones involving time. DataPacRat+1

Layer 1 only — Heim-inspired metronic geometry from area quanta 1) Primitive object (replace “bending plaquettes” entirely)

Each elementary unit is only:

an oriented area quantum

with fixed elementary area τ (metron-scale area)

no elastic internal modes

Mathematically:

$$\Sigma_\alpha \equiv (B_\alpha, \text{adj}(\alpha))$$

with

B_α an oriented simple bivector

$|B_\alpha| = \tau$ (fixed)

$\text{adj}(\alpha)$: combinatorial incidence to neighboring quanta

Important consequence

No “shape” (rectangle/triangle/hexagon) is fundamental anymore.

At Layer 1, shape is gauge/combinatorial representation.

You can realize the same bivector-quanta theory on triangular, square, or mixed cell complexes.

The invariant object is the oriented area element, not the polygon.

That said, for computation and constraints, triangles remain the cleanest discretization.

2) Dimension and coordinate principle (your revised axiom)

You want time initially treated the same as space. Good.

Axiom D1 — isotropic pre-physical manifold

Start with an n -dimensional metric-affine manifold (or discrete cell complex approximation) M^n , where no dimension is a priori designated as “time”.

Coordinates are initially just labels x^A , $A = 1, \dots, n$

Signature/asymmetry emerges later from admissible persistent curvature propagation

This avoids assuming Minkowski structure too early.

Axiom D2 — persistence requires temporal threading

A localized curvature excitation cannot exist as a single “spatial snapshot” only. It must extend as a connected chain/sheet through a distinguished cone-like causal direction once such a direction emerges.

This is exactly your statement:

for curvature to persist in space, it must also exist in the time dimension and move in the time cone.

In Layer 1 language, that becomes a topological/causal admissibility condition.

3) What is fundamental curvature if plaquettes do not bend?

If plaquettes are rigid and featureless, curvature cannot live “inside” a plaquette. It must live in relations between plaquettes.

So curvature is entirely encoded by:

adjacency

orientation mismatch

holonomy around closed loops

This is perfect for differential geometric algebra.

Curvature carrier = holonomy defect

Given a loop C in the adjacency graph (or dual complex), define a discrete transport operator $U(C)$. Then curvature exists iff:

$$U(C) \neq \text{Id}$$

for some minimal loop.

So the primitive curvature object is not “bending angle of a tile,” but a loop defect / holonomy bivector.

That matches your desire to remove plaquette bending while keeping geometry nontrivial.

4) Minimal geometric data structure (Layer 1 strict version)

Use a discrete complex K with cells and incidence maps.

Per area quantum α

Store only:

fixed norm bivector B_α , $|B_\alpha| = \tau$

orientation sign

incidence links to neighboring quanta

local frame label (or rotor to neighbor)

Per adjacency link $(\alpha\beta)$

Store:

a transport rotor / orthogonal map $R_{\alpha\beta}$

with compatibility $R_{\beta\alpha} = R_{\alpha\beta}^{-1}$

Per minimal loop C

Define holonomy:

$$H_C = \prod_{(\alpha\beta) \in C} R_{\alpha\beta}$$

Curvature is the loop defect H_C .

This is pure Layer 1. No energy, no wavefunction, no particles.

5) Constraints only (no dynamics yet)

You asked for geometric constraints only. Here they are.

C1. Quantized area constraint (metronic) $|B_\alpha| = \tau$

This is the discrete smallest-area postulate (Heim-like metron concept in secondary sources).
DataPacRat

C2. Simplicity constraint (true area element) $B_\alpha \wedge B_\alpha = 0$

This enforces that B_α is a simple bivector (represents a genuine 2-plane area element).

C3. Orientation consistency

If two quanta are glued along a shared boundary segment, induced orientations must be opposite on the interface.

(Discrete Stokes consistency.)

C4. Incidence/closure constraint

For a closed elementary boundary (e.g., around a 3-cell), oriented area contributions must close:

$$\sum_{\alpha \in \partial V} \epsilon_{\alpha V} B_{\alpha} = 0$$

This is a discrete analog of “boundary of boundary = 0” and local geometric closure.

C5. Metric compatibility of transport

Transport maps preserve area norm and grade:

$$R_{\alpha\beta} B_{\alpha} R_{\alpha\beta}^{-1} \in \Lambda^2, \quad |R_{\alpha\beta} B_{\alpha} R_{\alpha\beta}^{-1}| = \tau \quad \text{C6. Curvature localization by holonomy}$$

Curvature is only assigned to loops; no local loop defect means local flatness:

$$H_C = \text{Id} \quad \forall C \in \mathcal{L}_{\min} \Rightarrow \text{locally flat patch} \quad \text{C7. Causal persistence constraint (your key addition)}$$

A nontrivial curvature loop in a “spatial” slice must extend to a connected sequence of loop defects across neighboring slices in the emergent causal direction.

Informally: no isolated one-slice curvature events may count as persistent structures.

This is not quantum; it is a geometric persistence rule.

6) “Time equal to space” at the beginning, then distinguished later

This can be implemented without contradiction.

Phase A (pre-physical isotropic stage)

All dimensions treated uniformly

Curvature = holonomy defects of bivector quanta network

No time/space split yet

Phase B (causal foliation emerges)

A global consistency condition selects a cone structure (or effective signature), and one dimension-class becomes “temporal” operationally.

Then your statement becomes:

Persistent structures must form admissible chains/sheets inside the cone.

This is compatible with your photon remark:

if later one class of excitations corresponds to null-like persistence, proper-time accumulation may vanish (no internal elapsed proper time), but that belongs to Layer 2/3, so we stop here.

7) Re-checking Heim details (conservative summary only)

From the sources I checked (secondary summaries/overviews, not Heim’s primary German volumes), the following points align with your direction:

Heim-related descriptions emphasize a smallest area (“metron”) rather than infinitesimal continuum geometry. DataPacRat

The world is described as a quantized higher-dimensional geometric lattice, with local deformations corresponding to structures. DataPacRat+1

Secondary summaries classify structures by which dimensions participate (including cases involving time and extra dimensions), which supports your instinct that “existence in space” may require a structure in additional dimensions as well. DataPacRat

Auerbach/von Ludwiger also summarize Heim cosmology with an initial sphere-area criterion tied to metron size and expanding spheres, which matches your “sphere start + expansion” intuition (but we are postponing cosmology/energy here). DataPacRat

I’m being careful here: this is not a claim that Heim explicitly formulated your exact “causal persistence of curvature sheets” axiom. It is a mathematically natural Layer-1 formalization compatible with the broad metronic/higher-dimensional picture in those summaries.

8) Where GA/Hodge dual becomes useful already (still Layer 1)

You mentioned dual spaces / Hodge operator. Yes — even before quantum or phase-space talk.

Why Hodge is useful here

Given a metric and orientation (or emergent effective one), the Hodge dual maps:

$$\star : \Lambda^2 \rightarrow \Lambda^{n-2}$$

So each area quantum B_α has a dual $\star B_\alpha$.

This can simplify constraints:

closure constraints may be easier in dual variables (cochains)

holonomy defects can be represented either on primal loops or dual faces

causal persistence constraints may become linear incidence constraints in the dual complex

In practice:

primal: area quanta as bivectors

dual: flux-like cochains / codimension-2 carriers

That is likely the clean bridge to the “highly constrained movement / DoF reduction” you suspect — but still pure Layer 1.

9) Clean Layer-1 axioms (final version)

Here is a strict version you can adopt as the base theory.

Axiom L1 — Metronic area quanta

The substrate consists of elementary oriented area quanta B_α , each a simple bivector of fixed norm τ .

Axiom L2 — Combinatorial adjacency

Area quanta are organized in a finite/infinite cell complex with incidence relations and orientation signs.

Axiom L3 — No intrinsic plaquette dynamics

Area quanta possess no internal deformation modes; all geometric nontriviality resides in adjacency transport and loop holonomy.

Axiom L4 — Metric-compatible transport

Adjacent quanta are related by grade-preserving norm-preserving transport maps $R_{\alpha\beta}$.

Axiom L5 — Curvature as holonomy defect

Curvature is the nontrivial holonomy around minimal closed loops.

Axiom L6 — Local closure/compatibility

Every elementary cell satisfies discrete closure and orientation compatibility constraints.

Axiom L7 — Pre-physical dimensional symmetry

No coordinate direction is initially designated as time or space.

Axiom L8 — Persistent curvature admissibility

A localized curvature defect is a persistent structure only if it extends as a connected causal chain/sheet after emergence of a cone structure.

This is a coherent Layer-1 system.

10) Best next step (still no Layer 2/3)

I can next give you one of these (your choice, but I can also just do it directly):

- (A) Formal discrete equations in GA notation (bivectors, rotors, holonomy, closure)
- (B) A minimal 4D toy complex (3+1 treated symmetrically first) with explicit incidence matrices
- (C) A Grassmann.jl data model for $B\alpha$, $R\alpha\beta$, loops, and constraint checks only (no dynamics)
- (D) A dual-complex/Hodge formulation showing how closure + curvature constraints become sparse linear operators

Given your interests, (D + C) is probably the fastest route to something computable.

Sources
